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Systematic Review

Adoption of Agricultural Innovations Within the 'Farm to Fork' Strategy: A Realistic Review of Barriers, Paradoxes, and Avenues for Change

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Abstract

The transition towards sustainable agri-food systems, as envisioned in the European Union's Farm to Fork (F2F) Strategy, largely depends on the incorporation of technological innovations. However, existing literature has predominantly adopted confirmatory approaches focused on general benefits and barriers, without systematically addressing the conceptual and methodological tensions that arise in the implementation of policies of this magnitude. This study seeks to move beyond an incremental review, offering instead a critical and context-specific analysis. A realist review approach was employed to explore outcomes beyond average effects, guided by the central question: what works, for whom, under what conditions, and why? The selection of studies was based on conceptual relevance, including research on technological adoption within the framework of agricultural sustainability policies, even when explicit references to the F2F Strategy were absent. In addition, an epistemological quality scale was applied to weigh the evidence, distinguishing between different levels of methodological robustness, such as case studies and meta-analyses. The analysis shows that technologies such as precision agriculture and digitalization contribute to the objectives of F2F but also generate unforeseen tensions and contradictions during implementation. Conflicts emerge between environmental sustainability goals and short-term economic viability, especially among certain producer profiles. Barriers to adoption are not generic; rather, they vary according to a differentiated typology of small-scale producers, such as family farmers in peripheral EU regions or those transitioning to organic farming. Findings suggest that a strategy focused solely on technological promotion is insufficient. The main contribution of this study lies in the development of a conceptual framework to understand how technological adoption reshapes the tensions among the different pillars of the F2F Strategy, as well as the conditions under which innovation may hinder, rather than facilitate, the agroecological transition. The study concludes with policy recommendations advocating differentiated interventions tailored to the specific contexts of producers.

Keywords: realist review; technological adoption; precision agriculture; small-scale farmers; common agricultural policy; farm to fork strategy; agricultural innovation; sustainability; transition economies; global south; CMO analysis; adoption barriers

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1. Introduction

The transition to sustainable agri-food systems is one of the most pressing and complex challenges of the 21st century. In this context, the European Union has positioned the Farm to Fork (F2F) Strategy as a cornerstone for transforming the food system, setting ambitious goals that include reducing the use of chemical inputs, expanding organic farming, and optimizing the supply chain. Within this framework, technological innovations—from precision agriculture to digital monitoring systems—have been presented almost as a magic wand capable of fulfilling sustainability commitments.

The enthusiasm, however, contrasts with the quality of the available academic evidence. Much of the literature on technology adoption in agriculture has been limited to confirmatory and superficial exercise. Rogers' classic postulates on the diffusion of innovations have been validated time and again, and the usual socioeconomic barriers have been listed. But rarely has anyone investigated the unexpected tensions, the inevitable trade-offs, or the paradoxes that emerge when technology, instead of solving dilemmas, reconfigures them. This incremental approach, by reaffirming the obvious, contributes little to both the academic debate and the formulation of public policies with transformative capacity.

This study seeks to go beyond this descriptive level and offers a critical reading of the literature. Its contribution focuses on three areas: (1) exploring the intrinsic tensions that technological adoption can generate or intensify—for example, the friction between environmental sustainability and economic viability in the context of F2F; (2) applying a realistic review methodology that, through epistemological criteria, allows for the evaluation of the quality and relevance of the evidence, overcoming the limitations of overly restrictive approaches; and (3) proposing a conceptual framework that does not conceive of technology as a simple tool, but as a factor that shapes policy outcomes based on specific contexts, dismantling the abstract notion of a universal "small farmer."

The structure of the article follows this logic. Section 2 describes the methodology adopted, with special emphasis on the conceptual approach and critical evaluation of the evidence. Section 3 presents the findings, organized around the tensions identified and the paradoxes of innovation. Section 4 discusses the implications for policy design and develops the proposed conceptual framework. Finally, Section 5 presents the conclusions and suggests future lines of research.

2. Methods

A systematic literature review, following the exploratory review or scoping review approach [1], was conducted to analyze technological adoption in small-scale agriculture. This method is particularly suitable for mapping and synthesizing existing evidence, identifying knowledge gaps, and guiding future research [2]. The review was carried out in accordance with the PRISMA-ScR framework (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews) proposed by Tricco et al. (2018) [3]. The framework was adapted to ensure comprehensive and transparent coverage of the research field, incorporating multiple sources of information and maintaining rigor throughout the selection process [4,5].

The temporal scope focused on publications from the past five years (2019–2024), with the aim of capturing the most recent technological advances and their relevance for current policies, such as the "Farm to Fork" strategy [6]. An extensive search strategy was applied across specialized academic databases including Scopus, Web of Science (WoS), and SciELO, all of which are recognized for their broad coverage of scientific literature [7]. To ensure a global perspective with emphasis on transitioning economies, regional databases such as Redalyc and LATINDEX were also included, given their key representation of research in Latin America and Africa [8].

To maximize search accuracy, specific keywords were defined and combined using Boolean operators. Search terms included "technological adoption", "agricultural innovation," and "small-scale farmers". These terms were translated into Spanish and Portuguese to capture studies conducted in diverse linguistic and geographical contexts. Exclusion criteria focused on eliminating studies addressing large-scale producers or non-agricultural technologies, thereby ensuring that the findings and recommendations would be directly applicable to small-scale farmers.

2.1. Inclusion, Exclusion, and Quality Assessment Criteria

To guarantee the robustness and transparency of the review, a set of a priori criteria was established.

Inclusion criteria. Only peer-reviewed academic publications (empirical studies, case studies, systematic reviews, and meta-analyses) addressing the adoption of technologies and digital innovations by small-scale farmers, family farming, or small agricultural enterprises were considered. Eligible studies were those published between 2019 and 2024, available online in English, Spanish, or Portuguese.

Exclusion criteria. Editorials, conference abstracts, opinion-based contributions, and non-peer-reviewed reports were excluded. Research on technologies oriented towards large-scale producers or non-agricultural sectors, as well as publications prior to 2019 or without open access, were also discarded.

Quality assessment. A structured evaluation was applied through an epistemological quality scale, adapted from Moher et al. (2015) and Ayim et al. (2022) [9,10]. This scale combined methodological soundness, clarity of results, and relevance to the research questions, assigning scores from 0 to 3 in three domains:

1. Conceptual rigor (clarity of objectives, theoretical framing).
2. Methodological robustness (design adequacy, sampling, analytical techniques).
3. Relevance and contribution (alignment with agricultural innovation and adoption issues).

Studies scoring at least 6 out of 9 points were prioritized for inclusion. Greater weight was given to those indexed in Scopus or Web of Science, ensuring reliability and replicability. Disagreements between reviewers were resolved through consensus, with the intervention of a third reviewer when necessary.

2.2. Article Selection Process and Data Analysis

The study selection process followed the multi-phase structure recommended by the PRISMA-ScR framework [3]. An initial 3378 records were retrieved across five databases: Web of Science (n = 1684), Scopus (n = 1670), SciELO (n = 24), Redalyc, and Latindex. In addition, nine studies from the preliminary version of this review were incorporated to preserve continuity with prior analyses.

After duplicate removal (2312 records) and preliminary ineligibility screening performed with automated tools (200 records), the remaining 820 publications prior to 2019 were excluded. A total of 46 records advanced to title/abstract screening, from which 16 were excluded for thematic misalignment.

Subsequently, 30 full-text reports were retrieved for detailed evaluation. Nine reports could not be accessed, leaving 21 assessed for eligibility. Among these, 12 studies were excluded for methodological limitations, 8 for thematic misalignment with the Farm to Fork focus, and 1 due to poor presentation of results.

The final dataset incorporated 18 new eligible studies, which, combined with 9 from the previous review and 24 theoretical and conceptual contributions, resulted in a total of 51 studies included. This dataset forms the empirical and theoretical foundation for the subsequent analysis.

The selection process was carried out by three reviewers: the lead investigator conducted the initial screening of records, while two additional reviewers independently verified the adequacy and relevance of the included studies. Discrepancies were resolved by consensus, ensuring methodological transparency and reducing the risk of bias [11–13]. The complete flow of records through identification, screening, eligibility, and inclusion is depicted in Figure 1 (PRISMA flow diagram).

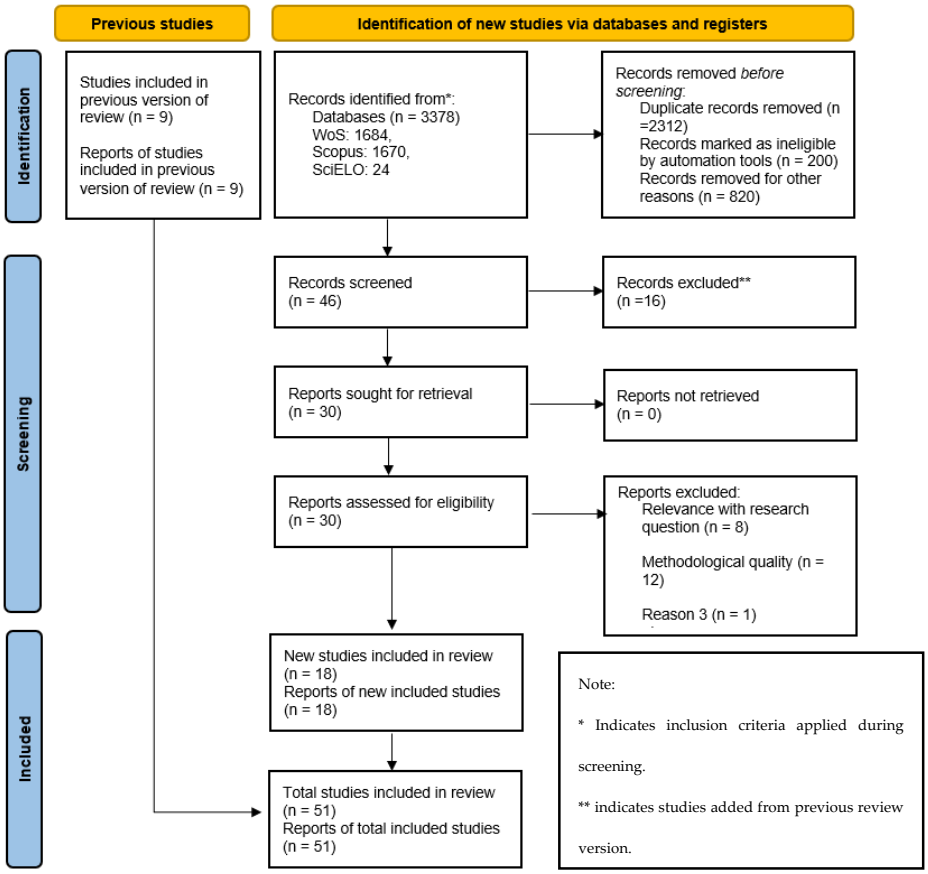


Figure 1. Presents the updated PRISMA flowchart illustrating the systematic selection and integration process of relevant studies on agricultural innovation and technology adoption among smallholder farmers. The diagram reflects a comprehensive multi-phase methodology: identification, screening, eligibility assessment, and inclusion.

This procedure aligns with the methodological recommendations of Liberati et al. (2009), Petticrew and Roberts (2006), and Wong et al. (2013), ensuring transparency and replicability [11–13].

2.3. Search Strategy and Keyword Mapping

The search strategy followed the systematic approach proposed by Tranfield et al. (2003) [14], ensuring broad and comprehensive coverage of the literature. Search equations were constructed by combining the terms "technological adoption", "agricultural innovation", and "small-scale farmers", complemented with equivalent terms in Spanish and Portuguese. Boolean operators and filters were applied to restrict results to the 2019–2024 period, peer-reviewed publications (empirical studies, case studies, systematic reviews, and meta-analyses), and three languages (English, Spanish, Portuguese).

The complete search strings, filtering criteria, and export logs for each database (Scopus, Web of Science, SciELO, Redalyc, and Latindex) are available in Supplementary Material S2, ensuring transparency and replicability.

In addition to the systematic search, a keyword co-occurrence analysis was performed during the first round of searches to identify thematic clusters and conceptual connections. The resulting network visualization is presented in Figure 2, which illustrates the frequency and interconnections of recurring terms. Three main clusters emerged from this analysis:

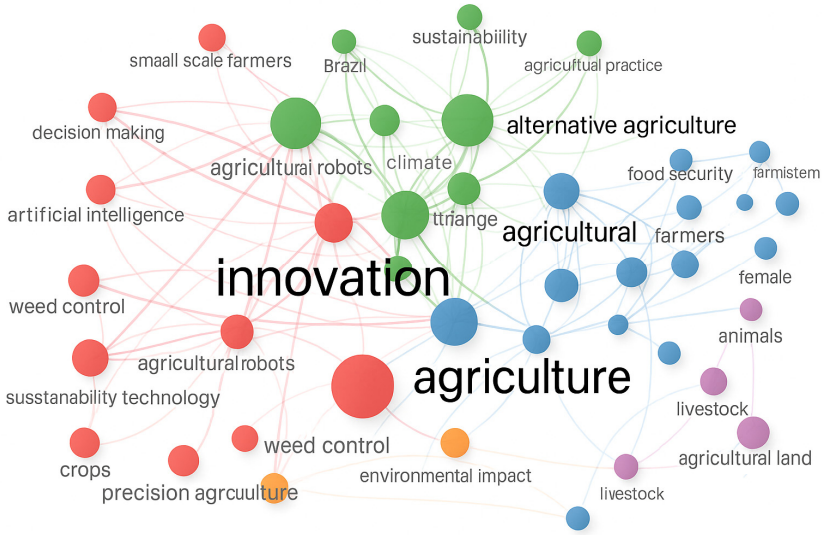


Figure 2. Keyword co-occurrence network of agricultural innovation and technology adoption (2019–2024). The visualization illustrates three thematic clusters: (1) technological innovations and impacts (blue); (2) sustainable practices and food security (green); and (3) development and efficient management (red). Node size represents frequency, while connecting lines indicate co-occurrence relationships.

Technological innovations and impacts (blue cluster). This cluster includes terms such as "advanced technology", "artificial intelligence", and "agricultural robots", underscoring their role in modernizing agriculture, improving efficiency, and reducing costs. Precision agriculture and environmental control systems exemplify how these innovations contribute to effective resource management and climate change mitigation [15,16].

Sustainable practices and food security (green cluster). Keywords grouped here include "sustainability", "alternative agriculture", and "environmental management", highlighting agricultural practices aimed at reducing environmental impacts while strengthening resilience and food security. These practices are crucial to ensure that production remains within ecological limits while enabling adaptation to changing conditions [17].

Development and efficient management (red cluster). This cluster focuses on issues such as land management, capital allocation, and farmers' decision-making processes. Strategic management of resources is essential to improve profitability and sustainability, while informed decisions regarding technology adoption directly influence productivity and long-term resilience. Farmers' attitudes toward innovation are therefore critical determinants of adoption effectiveness [18,19].

Together, these clusters provide valuable insights into the thematic density and interrelationships within the literature, complementing the systematic review process. By linking co-occurrence patterns with empirical evidence, the analysis offers a clearer understanding of how technological innovations, sustainable practices, and management strategies are transforming small-scale agriculture.

2.4. Eligibility and Quality Assessment Criteria

To ensure the rigor and transparency of the review, explicit inclusion and exclusion criteria were defined a priori. Eligible studies included peer-reviewed empirical articles, systematic reviews, bibliometric analyses, and conceptual contributions addressing the adoption of tech-

nologies and digital innovations by small-scale farmers, family farming, or small agricultural enterprises. Publications had to be available online between 2019 and 2024, in English, Spanish, or Portuguese. Exclusion criteria comprised editorials, opinion-based texts, conference abstracts, and studies focused on large-scale producers or non-agricultural technologies.

For the quality assessment, we applied an epistemological appraisal scale designed to prioritize studies with conceptual robustness and methodological transparency. This scale combined five complementary dimensions:

1. Clarity of research questions and objectives, ensuring alignment with the review scope.
2. Adequacy of study design, evaluating whether the methodology was appropriate for addressing technology adoption among small-scale farmers.
3. Validity and reliability of data collection and analysis, considering both internal and external validity.
4. Transparency and replicability, assessed through the availability of methodological details, instruments, or supplementary data.
5. Relevance to the Farm to Fork strategy and small-scale agriculture, highlighting contextual pertinence and policy implications.

Each study was assessed qualitatively according to these dimensions. Greater weight was assigned to publications indexed in Scopus and Web of Science, given their recognized reliability and peer-review standards [9]. Although no formal scoring system was applied, this multidimensional evaluation allowed us to filter studies that demonstrated sufficient methodological rigor and contextual alignment, thereby enhancing the robustness of the evidence base used in subsequent synthesis.

2.5. Data Items and Extraction Template

To ensure consistency and transparency in data handling, a standardized extraction template was developed and piloted prior to the systematic review. This template guided the collection of information from each study and allowed for the systematic organization of characteristics across the evidence base.

The following domains were extracted for every included study:

- *Bibliographic information*: author(s), year of publication, and citation details.
- *Geographical scope:* country or region where the study was conducted or to which it referred.
- *Study design*: type of study (empirical article, case study, bibliometric analysis, systematic review, or conceptual essay).
- *Population or sample*: small-scale farmers, family farms, agricultural enterprises, or aggregated publication corpora in bibliometric studies.
- *Technology or innovation:* type of agricultural practice or digital/technological tool analyzed (e.g., precision agriculture, CSA practices, digital platforms).
- *Theoretical framework*: models or perspectives employed (e.g., Diffusion of Innovations, institutional theory, realist CMO analysis).
- *Adoption of metrics and outcomes*: indicators such as adoption rate, yield, productivity, income, sustainability outcomes, or engagement.
- *Barriers and enablers*: factors influencing adoption, including costs, training, infrastructure, institutional support, or cultural constraints.
- *Funding and conflicts of interest*: explicit mention of research funding or reporting of independence.

All extracted information was systematically compiled in a master database (Supplementary Table S1), which includes the detailed characteristics of the 51 studies analyzed. This procedure ensured comparability across heterogeneous evidence and facilitated both thematic and realist synthesis described below.

2.6. Data Extraction Procedure

Data extraction was performed using a standardized and pilot-tested template to ensure consistency across studies. Two reviewers independently extracted the information, while the lead investigator supervised the process and verified the accuracy of the entries. Discrepancies were discussed and resolved through consensus, ensuring reliability and transparency.

The extraction template incorporated the domains previously defined (bibliographic information, geographical scope, study design, population, technology or innovation, theoretical framework, adoption metrics, barriers and enablers, and funding details). This dual-review process allowed us to maintain methodological rigor and minimize the risk of bias.

All extracted data were systematically compiled into a comprehensive matrix, available in Supplementary Table S1, which includes the full set of 51 studies analyzed. This matrix serves as the foundation for the thematic and realist synthesis presented in the following section.

2.7. Synthesis Approach

The evidence synthesis combined thematic analysis with a realist perspective, guided by the Context–Mechanism–Outcome (CMO) framework. This approach was chosen to capture not only the diversity of technological innovations in small-scale agriculture but also the specific contexts in which adoption occurs, the mechanisms that drive or hinder it, and the resulting outcomes.

Coding was carried out in two iterative phases. First, open coding was used to identify recurring concepts and categories across the extracted data, such as technological domains, institutional settings, and adoption barriers. Second, axial coding grouped these categories into broader themes, which were then aligned with the CMO framework to highlight how contextual conditions interact with mechanisms of adoption and generate specific outcomes. Evidence matrices and conceptual maps supported this process, ensuring traceability and transparency in the analytical pathway.

Given the heterogeneity of study designs, populations, and outcome measures, no quantitative meta-analysis was performed. Instead, a qualitative and realist synthesis was prioritized, which allowed the integration of empirical findings with theoretical insights and policy considerations. This decision follows established methodological recommendations for exploration and scoping reviews [20].

The combined thematic–realist synthesis provides a multidimensional understanding of the adoption of agricultural innovations, identifying common barriers and enablers while situating them within the policy framework of the Farm to Fork strategy.

3. Results

Our literature review identified a recurring pattern of low adoption of precision agriculture (PA) technologies among small-scale farmers. This finding is consistent across diverse geographical and socioeconomic contexts and is reflected in a persistent gap between the potential of PA for efficient resource management and its limited implementation in practice.

The literature reviewed indicates that this gap is not a matter of farmers' lack of interest but rather the result of systemic and contextual barriers. Empirical data from Smidt (2021) and Masere (2021) suggest that limited access to devices and insufficient technical training are key factors significantly restricting farmers' ability to effectively adopt PA technologies [21,22]. These findings are further supported by Restrepo-Campuzano et al. (2023), who reported that more than 50% of barriers to agricultural innovation are external in nature, linked to shortcomings in policy, infrastructure, and access to information [23].

3.1. Geographical and Temporal Distribution of Literature

The 51 studies included in this review display a heterogeneous geographical and temporal distribution, revealing both concentration and diversification processes in the field of agricultural technology adoption.

Geographically, the evidence base is strongly represented by countries with consolidated agricultural research systems such as Australia, Canada, New Zealand, South Africa, Sri Lanka, and the United Kingdom, each contributing multiple publications between 2019 and 2024. Their sustained output reflects not only the maturity of their innovative ecosystems but also their historical investment in agricultural research and extension infrastructures. Similarly, contributions from France, Spain, Greece, and the United States broaden the scope of perspectives, highlighting the role of high-income economies in shaping the global debate on technological adoption.

At the same time, studies from Latin America (Brazil, Mexico, Colombia, Argentina) and Africa (Ethiopia, Nigeria, South Africa) emphasize the growing participation of emerging economies, where technological adoption is often mediated by development programs, external funding, or private-sector initiatives. This regional diversity enriches the debate by providing context-sensitive insights that contrast with experiences in the Global North, addressing issues such as land fragmentation, financial constraints, and climate vulnerability, while highlighting the role of innovative technologies and international organizations in promoting sustainable agricultural development [24].

From a temporal perspective, research output has increased progressively since 2019, with a marked acceleration between 2021 and 2023. Early publications tended to be conceptual or exploratory, while more recent studies introduced empirical validation through econometric models, structural equation modeling, and policy evaluation. Notably, countries such as Australia, Canada, and China show consistent contributions across multiple years, suggesting that sustained policy support and innovation networks drive continuity in research agendas (Table 1).

Table 1. Geographical and temporal distribution of literature from the second round of the search in the SCR.

Country/Territory	Publications (n)	Years of Publication
Argentina	2	2021; 2024
Australia	3	2019; 2021; 2023
Bangladesh	2	2021; 2023
Brazil	3	2020; 2022; 2024
Canada	2	2020; 2024
China	3	2019; 2021; 2024
Colombia	2	2020; 2023
Ethiopia	4	2021; 2022; 2023; 2024
France	1	2021
Greece	1	2020
India	3	2020; 2022; 2023
Mexico	3	2019; 2022; 2023
New Zealand	2	2021; 2022
Nigeria	2	2020; 2024
South Africa	2	2020; 2023
Spain	2	2022; 2023
Sri Lanka	1	2019
United Kingdom	3	2019; 2022; 2024
United States	2	2021; 2024
Global/Other	18	2019–2024 (multiple years)
Total	51	—

Taken together, the geographical and temporal distribution underscores two complementary dynamics. On the one hand, leadership remains concentrated in countries with established infrastructures and innovation systems; on the other, the expansion of studies in the Global South demonstrates a gradual diversification of perspectives, which is essential for understanding the differentiated realities of small-scale farmers worldwide. This combined distribution provides the empirical foundation for subsequent sections, where specific typologies of farmers and multidimensional adoption factors are analyzed.

3.2. Results of the Bibliometric Analysis

The bibliometric analysis offers a quantitative assessment of the current research landscape on technology adoption among smallholder farmers. This approach identifies key trends, publication patterns, and the geographical and temporal distribution of existing literature [25]. By understanding how the field has evolved recently and identifying the primary areas of research, this analysis provides a strong foundation for developing new research directions and innovation strategies in agriculture [26].

First Round of Scope Review Search

The bibliometric analysis provides a quantitative overview of the research landscape on agricultural technology adoption among small-scale farmers, highlighting both the scope of available evidence and the rigor applied in filtering studies to ensure relevance and quality.

As shown in Table 2, a total of 3378 records were initially retrieved from international and regional databases. Web of Science (WoS) contributed 1684 records (49.9%), Scopus 1670 records (49.4%), and SciELO 24 records (0.7%). This distribution confirms the dominance of WoS and Scopus as the most comprehensive sources of peer-reviewed literature in the field, while regional platforms such as SciELO provided only marginal contributions. The decision to combine international and regional databases was therefore validated, ensuring a balanced perspective between high-impact outlets and region-specific research [25,26].

The screening process progressively reduced the dataset. First, 2312 duplicate records (68.4%) were removed. Then, 820 records (24.3%) published before 2019 were excluded to maintain temporal relevance to contemporary agricultural technologies and the Farm to Fork strategy. Additionally, 200 entries (5.9%) were flagged as ineligible by automated tools based on scope and formatting criteria. After these filtering stages, 46 records (1.4%) proceeded with detailed screening.

From these, 36 studies (1.1%) were excluded for thematic misalignment, as they did not explicitly address small-scale farmers or the Farm to Fork framework. The remaining 30 full-text reports (0.9%) underwent eligibility assessment. Of these, 12 were discarded for methodological limitations, 8 for lack of direct relevance to the research questions, and 1 for inadequate result presentation. Consequently, only 9 studies (0.27%) were incorporated from this first-round review.

While the inclusion rate appears low, this reflects the stringency of the methodological criteria applied and underscores the relative scarcity of directly relevant empirical evidence. To compensate for this limitation, an additional 24 conceptual and theoretical contributions and 18 new empirical studies identified during the refinement phase were incorporated. Together, these sources provided a total of 51 studies, which constitute the final evidence base for the present review.

This selective process ensured that the included studies are both methodologically sound and directly aligned with the objectives of the review, providing a robust foundation for subsequent thematic and realist synthesis.

Table 2. Distribution of records and studies throughout the bibliometric screening and selection process (2019–2024).

Stage of Screening and Inclusion	Records (n)	% of Initial Dataset
Records identified (WoS, Scopus, SciELO, Redalyc, LATINDEX)	3378	100
Duplicate records removed	2312	68.4
Records excluded before 2019	820	24.3
Records marked ineligible by automated tools	200	5.9
Records screened by title/abstract	46	1.4
Records excluded (thematic misalignment)	36	1.1
Reports assessed for eligibility (full text)	30	0.9
Reports excluded for methodological limitations	12	0.35
Reports excluded for lack of relevance	8	0.24
Reports excluded for poor presentation	1	0.03
Studies included from first-round review	9	0.27
Additional conceptual/theoretical studies	24	—
New eligible studies (second round)	18	—
Final studies included in the review	51	—

3.3. Second Round of Review

To overcome the tendency to treat small-scale farmers as a homogeneous group, this review proposes a typology that differentiates them according to their socioeconomic and institutional contexts. This approach enhances analytical precision and provides a stronger basis for deriving targeted policy implications. Three main groups are identified:

(A) Small-Scale Farmers in the European Union (EU).

These farmers operate under the framework of the Common Agricultural Policy (CAP), which strongly influences their production and adoption decisions. However, CAP benefits are distributed asymmetrically: 80% of direct payments are concentrated among 20% of beneficiaries, while farmers with less than five hectares receive only 4% of total subsidies [27]. As a result, many small-scale EU farmers depend critically on subsidies for survival and face unequal competition with larger producers. This structural imbalance not only limits their autonomy but also conditions the feasibility of adopting innovative or sustainable practices.

(B) Farmers in Transition Economies.

Producers in Eastern Europe and former Soviet republics have been shaped by processes of privatization and fragmentation after the collapse of collective systems. The resulting small plots and insecure tenure arrangements, combined with limited access to credit and weak institutional frameworks, constrain their ability to adopt new technologies [28]. In this context, barriers are less related to willingness and more to structural conditions: lack of financing, absence of extension services, and fragmented market integration.

(C) Subsistence Farmers in the Global South.

Located mainly in Africa, Latin America, and parts of Asia, these farmers primarily produce for household consumption and survival. Their main priorities are food security and resilience to climatic shocks. Limited access to land tenure security, microcredit, technology, and formal markets exacerbate their vulnerability. Reports by FAO (2014) and the Inter-American Development Bank (IDB, 2021) highlight how these producers often face high exposure to poverty and hunger, while lacking institutional mechanisms to support sustainable adoption [29,30]. In this setting, psychosocial and adaptive capacities play as decisive a role as economic incentives.

This typology is summarized in Table 3, which highlights the main characteristics and barriers affecting small-scale farmers across the European Union, transition economies, and the Global South.

Table 3. Typology of Small-Scale Farmers at the Global Level.

Context	Main Characteristics	Predominant Barriers
European Union (EU)	Operate within the Common Agricultural Policy (CAP). Small plots often <5 ha. Strong dependence on subsidies for viability.	Unequal subsidy distribution (80% of payments go to 20% of farmers). Competitive disadvantage against larger farms. Limited incentives for innovation.
Transition Economies	Producers in Eastern Europe and former Soviet republics. Farms fragmented after privatization of collective land. Weak institutional frameworks.	Land tenure insecurity. Lack of credit and financing mechanisms. Insufficient extension services. Poor market integration.
Global South (Africa, Latin America, Asia)	Subsistence-oriented farming. Focus on household food security. High exposure to climatic shocks.	Limited land tenure security. Lack of microcredit and access to technology. Informal or absent market channels. High vulnerability to poverty and hunger.

This layered perspective reveals that while EU farmers struggle with subsidy dependence, producers in transition economies face institutional voids, and Global South subsistence farmers contend with extreme vulnerability. Recognizing these distinctions is essential for designing differentiated policy instruments, as the challenges and opportunities for adoption vary substantially across contexts.

Table 4 summarizes the primary studies addressing technological innovation and sustainability in the agri-food sector.

Table 4. Overview of primary studies addressing technological innovation, sustainability, and technology adoption in the agri-food sector, with emphasis on small producers and supply chains.

Sl	Author	Title of the Document	Year	Summary
1	Ponnampalam E.N.; Bekhit A.E.D.; Bruce H.; Scollan N.D.; Muchenje V.; Silva P.; Jacobs, J.L. [31]	Production Strategies and Meat Processing Systems: Current State and Future Prospects for Innovation—A Global Perspective	2019	Research in nutrition, genetics, animal welfare, meat production and human health has advanced
	Paarlberg R. [32]	The transatlantic conflict over “green” agriculture	2022	With its new Farm to Fork (F2F) strategy, the EU plans to expand organic farming, an approach that excludes both synthetic chemicals and modern biotechnology
2	Martínez-Castañeda M.; Feijoo C. [33]	Use of blockchain in the agri-food value chain: State of the art in Spain and some lessons from the perspective of public support.	2023	The European Union’s Common Agricultural Policy (CAP) seeks to differentiate products in terms of quality, providing greater transparency on the origin of food.
	Caro, Miguel P., Muhammad Salek Ali, Massimo Vecchio, y Raffaele Giaffreda. [34]	Blockchain-based traceability in agri-food supply chain management: a practical implementation	2018	The recent exponential increase in the adoption of the most disparate Internet of Things (IoT) devices and technologies has also reached agricultural supply chains

Table 4. *Cont.*

Sl	Author	Title of the Document	Year	Summary
3	Pincheira M.; Vechio M.; Giaffreda R. [35]	Exploiting Cost-Effective IoT Devices for Trustless Agri-Food Supply Chain Management: A Case Study	2022	The exponential increase in the adoption of various Internet of Things (IoT) devices has reached the supply chains of Agriculture and Food (Agri-food).
4	Raptou E.; Mattas K.; Tsakiridou E.; Baurakis G. [36]	Assessing the aftermath of the COVID-19 outbreak on the agri-food system: an exploratory study of expert perspectives.	2022	The present study explored the impacts of the COVID-19 outbreak on the food system in terms of agri-food production and efficiency of distribution networks.
5	Ponnampalam E.N.; Bekhit A.E.D.; Bruce H.; Scollan N.D.; Muchenje V.; Silva P., J.L. [31]	Meat Production Strategies and Processing Systems: Current State and Future Prospects for Innovation: A Global Perspective	2018	Research in nutrition, genetics, animal welfare, meat production and human health has advanced in parallel with technological advances
6	Bellon-Maurel V.; Piot-Lepetit I.; Lachia N.; Tisseyre B. [37]	Digital agriculture in Europe and France: which organizations can drive adoption levels?	2023	This article presents how the digital transformation of the agricultural sector is being implemented in Europe and in France.
7	Bucci G.; Bentivoglio D.; Finco A. [38]	Precision agriculture as a driver of sustainable agricultural systems: state of the art in literature and research	2018	In 2017, the food and beverage industry was confirmed as the largest manufacturing sector in the European Union, in terms of employment, turnover and value added.
8	Rijswijk, K; Klerkx, L; Bacco, M; Bartolini, F; Bulten, E; Debruyne, L; Dessein, J; Scotti, I; Brunori, G [39]	Digital Transformation of Agriculture and Rural Areas: A Socio-Cyber-Physical System Framework to Support Accountability	2021	Digital technologies are often seen as an opportunity to enable sustainable futures in agriculture and rural areas. However, this process of digital transformation is not inherently good.
9	Metta, M; Dessein, J; Brunori, G [40]	Between the on-site and the cloud: Socio-cyber-physical assemblages in on-farm diversification	2024	This paper sheds light on the integration of digitalization into multifunctional and diversified agriculture.
10	Westerlund, M; Nene, S; Leminen, S; Rajahonka, M [41]	An Exploration of Blockchain-Based Traceability in Food Supply Chains: On the Benefits of Distributed Digital Records from Farm to Fork	2021	There are increasing internal and external pressures for traceability in food supply chains due to food scandals.
11	Righi, S; Viganò, and [42]	How to ensure the sustainability of organic food system farms? Environmental protection and fair price	2023	With the implementation of the Farm to Fork Strategy, the European Union aims to drastically reduce the use of synthetic chemical inputs
11	Heyl, K; Ekardt, F; Sund, L; Roos, P [43]	Potentialities and Limitations of Subsidies in Sustainability Governance: The Example of Agriculture	2022	The goals of the Paris Agreement and the Convention on Biological Diversity call for a global transition to sustainability.
12	Tensi, AF; Ang, F; van der Fels-Klerx, HJ [44]	Microbial Applications and Agricultural Sustainability: A Simulation Analysis of Dutch Potato Farms	2024	Fertilizers and plant protection products are essential for the economic viability of arable agriculture, but their overuse causes environmental problems.
13	Drechsel, P; Qadir, M; Galibourg, D [45]	a review of implementation challenges and potential solutions in the Global South	2022	Globally, untreated, often diluted or partially treated wastewater is used in agriculture.
14	Lyse, F; Manikas, I; Apostolidou, I; Wahbeh, S [46]	The role of traceability in end-to-end circular agri-food supply chains	2022	The transition to a circular supply chain is a prerequisite for the agri-food sector to be able to cope with the growing consumer pressure for sustainability, while meeting the required quality and safety standards.

3.4. Multidimensional Factors in the Adoption of Sustainable Agricultural Technologies (CSA)

The adoption of sustainable agricultural technologies within the framework of Climate-Smart Agriculture (CSA) has been analyzed through diverse theoretical lenses, underscoring its multidimensional nature. Economic constraints, institutional frameworks, psychosocial perceptions, and contextual vulnerabilities converge to shape farmers' decisions and outcomes.

Teklewold et al. (2019), in a study conducted in Ethiopia, propose an integrative model combining economic theory, innovation diffusion, and perception analysis [47]. Their multivariate Probit results demonstrate that access to credit, the number of agricultural plots, and participation in social networks significantly influence the simultaneous adoption of multiple CSA practices, pointing to the complementarities among technologies.

In rural China, Wang et al. (2024) employ logistic regression and structural equation modeling (SEM) to assess institutional drivers of CSA adoption [48]. They show that access to technical services, the availability of climate information, and trust in agricultural institutions are decisive determinants that increase the probability of transitioning to sustainable practices. These findings emphasize the centrality of institutional support in enabling adoption.

From a behavioral perspective, Belachew et al. (2024) examine how exposure to extreme climate events shapes farmers' perceptions of risk. Their survey-based evidence in Ethiopia suggests that higher climate-risk awareness translates into a stronger willingness to adopt CSA technologies, highlighting the importance of psychological and adaptive factors in vulnerable environments [49].

Complementing these perspectives, Danso-Abbeam et al. (2021) examine the socioeconomic impacts of non-farm employment on climate-change adaptation among Nigerian smallholders [50]. Using an endogenous treatment Poisson model complemented by IPWRA estimation, they find that participation in non-farm activities significantly enhances farmers' adaptive capacity and household welfare, reinforcing the link between sustainability, productivity, and poverty reduction.

Together, these studies confirm that CSA adoption cannot be explained by economic or technological variables alone. Instead, it emerges from the interplay of structural, institutional, psychosocial, and economic dimensions, offering a robust framework for comparative analysis (Table 5).

Table 5. Empirical Studies on CSA Adoption: Methods, Theoretical Frameworks, and Key Findings.

Authors and Year	Country	Method	Theoretical Framework	Key Finding
Teklewold et al. (2019) [47]	Ethiopia	Multivariate Probit	Economics + Innovation Diffusion + Perception	Credit access, number of plots, and social networks influence multiple CSA adoption.
Wang et al. (2024) [48]	China	Logistic regression + SEM	Institutional and Structural	Institutional trust and technical assistance increase CSA adoption.
Belachew et al. (2024) [49]	Ethiopia	Survey and risk perception analysis	Risk psychology	Perceived climate risks promote adoption of CSA technologies.
Danso-Abbeam G., Ojo T.O., Baiyegunhi L.J.S., & Ogundeji A.A. (2021) [50]	Nigeria	Endogenous Treatment Poisson Model comple-mented by In-verse-Probability-Weighted Regression Adjustment (IPWRA)	Household utility maximiza-tion and time-allocation framework linking non-farm employment to adaptive capacity	Participation in non-farm employment significantly enhances smallholders' adaptive capacity by increasing the number and diversity of climate-change adaptation strategies, thereby improving household welfare and resilience.

Figure 3 illustrates the temporal evolution of academic publications on agricultural innovation and technology adoption from 2003 to 2025. The data reveal a steady growth in interest after 2019, with a peak in 2021 (8 studies) and sustained production in 2022 (7 studies) and 2023–2024 (6 studies each). This upward trend aligns with the acceleration of sustainability-oriented frameworks such as the European Green Deal and the Farm to Fork strategy, as well as the post-pandemic focus on agricultural resilience. The figure supports the methodological decision to prioritize studies published after 2019, ensuring that the review reflects the most updated and policy-relevant evidence base.

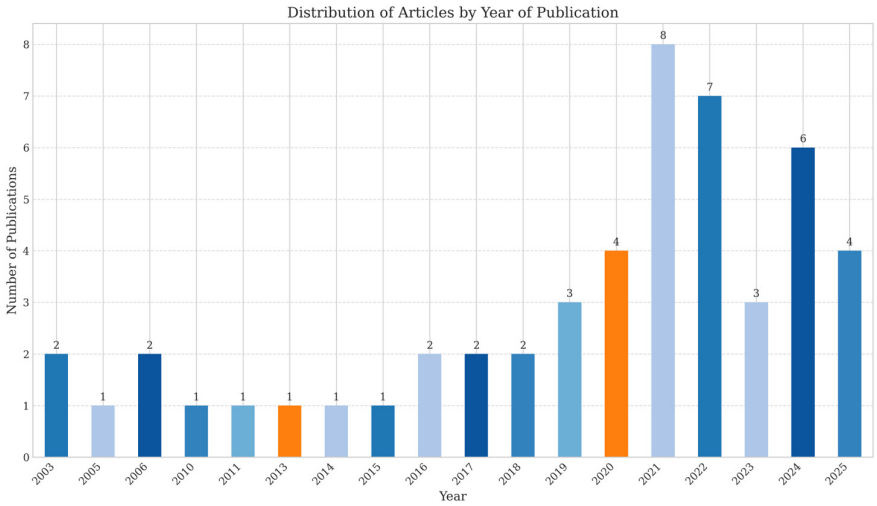


Figure 3. Temporal Evolution of Academic Publications on CSA (2003–2025).

This multidimensional perspective provides a valuable framework for examining technology adoption in contexts such as Colombia, where rural producers face structural fragmentation, institutional limitations, and high vulnerability to climate variability. Integrating economic, institutional, and psychosocial dimensions is therefore essential for designing effective interventions tailored to the realities of small-scale farmers.

3.5. Recent Contributions on Sustainable Approaches and Perceptions in Technology Adoption

Recent studies have expanded the literature on sustainable agricultural technology adoption by incorporating interdisciplinary perspectives that move beyond classical economic and institutional models. For instance, Negera et al. (2022) analyze the determi-nants of multiple climate-smart agricultural practices among Ethiopian smallholders, highlighting the role of extension frequency, education, land size, and household assets in enhancing adoption and intensity of CSA use [51]. Similarly, Wang et al. (2024) demonstrate how institutional trust and access to extension services increase the likelihood of Chinese farmers adopting climate-smart agriculture practices [48].

From a psychological and educational perspective, Jabbar et al. (2022) show that participatory approaches and Farmer Field Schools (FFS) strengthen farmers' knowledge and perceptions of sustainability and increase the adoption of sustainable practices [52]. Complementarily, Ruzzante et al. (2021) provide global meta-analytic evidence that hu-man-capital and institutional channels—such as education, access to extension services, credit, land tenure, and organization membership—are positively associated with agri-cultural technology adoption in developing countries [53]. Complementarily, Aguerre & Bonina (2024) introduce digital literacy and social media use as explanatory variables, revealing how online platforms and virtual communities facilitate the diffusion of knowledge and the acceptance of innovation across rural areas in Latin America [54] (Table 6).

Table 6. Summary of recent studies on sustainable technology adoption in agriculture.

Author (s) and Year	Country/Region	Methodology	Focus Area	Key Findings
Negera et al. (2022) [51]	Ethiopia	Multivariate Probit and Ordered Probit Models	Adoption of multiple Cli-mate-Smart Agricultural (CSA) practices	Education level, landholding size, access to extension ser-vices, livestock ownership, and farm income significantly increase both the likelihood and intensity of CSA adoption among smallholder farmers.
Wang et al. (2024) [48]	China	SEM + Logit models	Institutional drivers of CSA	Access to extension and institutional trust enhance adoption probability.
Jabbar et al. (2022) [52]	Pakistan	Recursive Bivariate Probit (RBP) and Propensity Score Matching (PSM)	Impact of Farmer Field Schools (FFS) on sustaina-ble agricultural practices and productivity	Participation in FFS enhances farmers' knowledge, promotes adoption of sustain-able agricultural practices, and increases citrus yield among smallholders.
Ruzzante et al. (2021) [53]	Global (Me-ta-analysis of developing countries)	Meta-analysis of 320 empirical studies	Determinants of agricultural technology adoption	Human-capital and institutional factors—education, extension access, credit, and land tenure—consistently drive higher adoption rates of agricultural technologies in developing countries.
Aguerre & Bonina (2024) [54]	Latin America	Mixed methods	Digital adoption and social media use	Online platforms and digital literacy facilitate knowledge diffusion and foster the adoption of sustainable innovations.

These contributions reinforce the need for multidimensional frameworks that account not only for structural and institutional conditions but also for psychosocial and digital factors. The integration of risk perception, participatory education, and digital adoption into the analysis of small-scale farmers provides a more comprehensive understanding of the adoption process. Such approaches will be crucial for designing culturally sensitive and context-specific interventions that foster both sustainability and resilience in agricultural systems.

3.6. Thematic Classification of Literature

To provide greater clarity in the analysis, the reviewed studies were thematically classified into four main categories that capture the dominant research perspectives: (i) Technology & Innovation, (ii) Environmental Sustainability, (iii) Social and Institutional Factors, and (iv) Data & Research Approaches. This classification synthesizes the diverse approaches in literature, enabling a more systematic understanding of how different factors shape the adoption of agricultural innovations.

The first category, Technology & Innovation, encompasses studies focusing on the development and application of new tools, including precision agriculture, digital platforms, and disruptive technologies such as IoT and blockchain [33,55]. These works highlight how technological advances contribute to improved efficiency and market integration.

The second category, Environmental Sustainability, includes research analyzing how technological adoption addresses ecological challenges, climate change, and food security. Studies such as Righi et al. (2023) and Belachew et al. (2024) emphasize the dual objective of maintaining productivity while reducing environmental impact [42,49].

The third category, Social and Institutional Factors, addresses the influence of governance, subsidies, and community dynamics on adoption decisions. Examples include Heyl et al. (2022)

on the role of subsidies in sustainability governance and Wang et al. (2024) on the importance of institutional trust and extension services [43,48].

Finally, the category Data & Research Approaches groups studies that advance methodological frameworks for systematic reviews and empirical analysis, such as Tricco et al. (2018) with the PRISMA-ScR guidelines and Higgins & Green (2011) with methodological handbooks [3,56]. These contributions establish rigorous standards that strengthen the validity and comparability of research outcomes.

This thematic classification, supported by the conceptual framework in Figure 4, illustrates how technological, environmental, social, and methodological dimensions interact in shaping the adoption of agricultural innovations. By organizing the literature in this way, the review not only highlights persistent asymmetries but also reveals opportunities for interdisciplinary approaches that integrate technical, institutional, and behavioral insights.

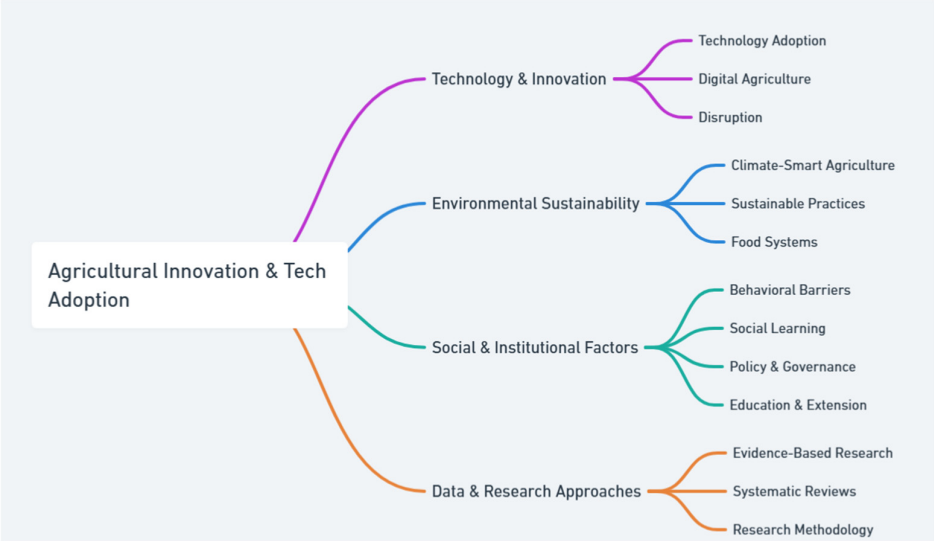


Figure 4. Framework of Factors Influencing Agricultural Innovation and Technology Adoption.

4. Discussion

4.1. Proposed Theoretical Perspective

The studies reviewed indicate that implementing technologies in the agricultural sector is a complex process influenced by socioeconomic, demographic, and institutional factors. Rogers' (2003) diffusion of innovations theory posits that technology adoption depends on the innovation's characteristics, perceived usefulness, and the social context of the user [57]. This theory is evident in the results, which show that smallholder farmers, particularly in developing regions, face limitations such as lack of infrastructure, institutional support, and training necessary to adopt advanced technologies, as reported by Dibbern et al. (2024) [58]. Furthermore, Chindasombatcharoen et al. (2024) highlights that adoption is also hindered by internal psychological barriers—such as low trust, perceived complexity, and resistance to change, which reinforces the idea that non-material and behavioral factors are critical to understand innovation resistance among smallholders [59].

4.1.1. Precision Agriculture and Barriers to Adoption

The low adoption of precision agriculture (PA) technologies, presented in the Results Section 3, goes beyond a simple description and can be explained through the Realist Review (CMO) framework. This methodology allows us to unravel the underlying logic that connects contextual factors with observed outcomes, providing a causal explanation rather than a mere

correlation. Such an approach is crucial to understanding the complexity of technological adoption in small-scale agriculture.

The lack of adoption of precision agriculture is not an inherent flaw of technology but rather a direct result of mechanisms operating within specific contexts.

Context (C): Farmers in developing countries operate in environments characterized by limited digital infrastructure and scarce access to financial resources. Rogers' (2003) theory highlights that successful technological implementation requires contexts that provide adequate resources and institutional support. This aligns with the findings of Smidt (2022) and Masere (2021), who emphasize the shortage of devices and technical training as key barriers [21,22,57].

Mechanism (M): The convergence of contextual barriers triggers a mechanism that restricts farmers' agency. Although precision agriculture could enhance efficiency, the lack of access to financing and technical information generates a cycle of inaccessibility and distrust. In this scenario, innovation is not perceived as a viable solution but rather as a risk or an unattainable investment.

Outcome (O): The result is low or even null adoption of PA technologies. This outcome not only limits the potential for improvements in productivity and resource management but also perpetuates existing inequalities. The findings of Restrepo-Campuzano et al. (2023) reinforce this point, indicating that external barriers (such as deficiencies in infrastructure and policy) are the main drivers of low adoption, suggesting that improved regulations and incentives are essential to catalyze change [23].

4.1.2. Disparities in the Development of Digital Agriculture

Research reviews demonstrate that progress in digital agriculture has been more pronounced in developed countries. Ciruela-Lorenzo (2020) highlights how robust infrastructure and financing in these countries have facilitated the effective and accelerated implementation of advanced technologies [60]. This disparity underscores a limitation in the diffusion of innovations theory when applied to contexts with limited resources. Paarlberg's (2022) policy analysis suggests the need for public policies that promote equitable access to digital technologies, ensuring that smallholders in resource-constrained regions can also benefit from technological advancements [32]. As Weber et al. (2025) emphasize, the EU's Farm to Fork strategy—although essential for long-term sustainability—could reduce agricultural output and increase import dependency if it is not paired with innovation-driven solutions [61].

4.1.3. Contributions to the Sustainable Development Goals (SDGs)

The integration of digital technologies in agriculture aligns with several Sustainable Development Goals (SDGs), particularly climate action and sustainable resource management. Studies like Kamble, Gunasekaran & Sharma (2020) demonstrate that technologies like blockchain enhance traceability and sustainability in supply chains, increasing transparency and consumer confidence [62]. These benefits support SDG targets, indicating that technology adoption can serve as both an economic advantage and a critical component for global sustainability. Furthermore, the European Green Deal and Regulation (EU) 2019/1009 emphasize sustainability criteria that directly relate to agricultural technology compliance, reinforcing the role of digital tools in achieving climate neutrality and food safety across regions [63].

4.1.4. Integration of Science and Practical Experience

Adopting technologies in agriculture requires both scientific knowledge and practical experience. Siva's research on drip irrigation in India illustrates that technology can improve water efficiency and productivity in water-scarce contexts. Mohan, G. (2024) recommends conducting practical experiments to adapt technologies to local conditions, ensuring effective

implementation [64]. Comparative analysis across different agricultural contexts suggests that successful technology adoption must transcend mere availability of tools. Public policies must ensure equitable access to technologies, considering the economic, social, and environmental conditions of each region to foster inclusive and sustainable sector development.

Moreover, developing training and technical monitoring programs that support farmers is essential for the effective use of innovative tools. Training should focus on promoting the practical and localized application of innovations, integrating them seamlessly with existing agricultural practices. This approach enhances productivity and sustainability and enables small farmers to adopt technologies and actively participate in their productive transformation, thereby promoting resilient and efficient resource management in agriculture.

4.1.5. Institutional Support and Multi-Actor Governance

In addition to individual and technological factors, institutional and multi-actor collaboration plays a critical role in shaping technology adoption pathways. Pedersen et al. (2024) emphasize that barriers such as lack of financial incentives, misaligned policies, and limited coordination between stakeholders hinder the transition toward climate-smart agriculture in the EU [65]. Their findings underscore the importance of combining top-down governance with bottom-up engagement. Likewise, Restrepo-Campuzano et al. (2023) advocate for participatory mechanisms and tailored regulatory instruments to overcome the most persistent external barriers [23]. These insights suggest that innovation policies should be context-sensitive, farmer-centered, and supported by inclusive governance mechanisms.

4.2. Paradoxes and Contradictions in Small-Scale Agriculture

The analysis of small-scale agriculture reveals paradoxes that illustrate the tensions between sustainability goals and the structural realities faced by farmers. One of the most prominent is the organic agriculture yield paradox. While organic practices are promoted as pathways toward environmentally friendly production, empirical studies consistently show that they often result in lower yields compared to conventional methods [66]. This yield gap generates a contradiction: policies encouraging organic farming contribute to sustainability narratives, but in practice they risk undermining food security and farmers' income. Exploring Solutions: Evidence from FAO (2014) and Teklewold et al. (2019) suggests that yield reductions can be mitigated through mixed production systems, precision nutrient management, and targeted subsidies that compensate for short-term productivity losses [29,47]. Pilot programs in the European Union under the Farm to Fork strategy, as well as Latin American experiences in agroecological transitions, indicate that integrating organic methods with digital monitoring tools and climate-smart practices can narrow the productivity gap without sacrificing environmental goals.

A second paradox is the well-intentioned regulation paradox. Regulations designed to ensure food safety, traceability, and sustainability often impose administrative and financial burdens that small-scale farmers are least equipped to manage [43]. This generates double pressure: the rules aim to protect consumers and ecosystems but inadvertently exclude smallholders from formal markets. Exploring Solutions: Studies by IDB (2021) and López Martínez (2021) highlight that simplifying certification procedures, subsidizing compliance costs, and promoting collective certification schemes through cooperatives can reduce these barriers [30,67]. Empirical evidence from India's participatory guarantee systems and Colombia's associative export platforms demonstrates that shared compliance mechanisms and digital traceability tools adapted for low-resource settings significantly reduce transaction costs and facilitate inclusion of small-scale producers in higher-value chains.

Finally, the innovation paradox emerges when new technologies such as digital platforms, sensors, and blockchain systems are introduced as enablers of sustainability but remain largely inaccessible to small-scale farmers due to cost, connectivity gaps, and lack of digital literacy [54]. This results in a widening digital divide, where the same technologies that could empower smallholders, risk reinforcing structural inequalities. Exploring Solutions: Evidence from Wang et al. (2024) in China and Belachew et al. (2024) in Ethiopia indicates that access to extension services, institutional trust, and farmer training play a decisive role in bridging this divide [48,49]. Community-based digital hubs and shared service platforms, piloted in Sub-Saharan Africa, show that collective access models combined with microcredit and targeted capacity-building programs can make advanced technologies viable for smallholders. Integrating these tools with climate-smart agriculture frameworks reinforces both technological adoption and resilience to climate risks.

Together, these paradoxes highlight the critical need for context-sensitive interventions. Addressing them requires aligning policy frameworks with the realities of small-scale farmers, leveraging targeted support measures, and promoting adaptive innovations that minimize trade-offs between productivity, equity, and sustainability. By situating instinctive solutions within existing empirical experiences, from EU subsidy reforms to participatory schemes in the Global South, the discussion underscores that paradoxes are not insurmountable, but rather opportunities to redesign agricultural policies toward inclusivity and resilience.

4.3. Theoretical Contributions

This review advances theoretical debates on technology adoption in agriculture by clarifying the distinction and complementarities between classical diffusion frameworks and realist approaches. The diffusion of innovations model proposed by Rogers (2003) has been widely used to explain how new technologies spread across social systems, emphasizing categories of adopters (innovators, early adopters, majority, laggards) and the role of communication channels in accelerating or slowing diffusion [57]. While this perspective has provided an essential baseline for understanding agricultural innovation processes, its linear and universalist assumptions often fail to capture the contextual variability that characterizes small-scale farming systems.

In contrast, the realist review approach, operationalized through the Context–Mechanism–Outcome (CMO) framework, focuses on explaining not only whether an innovation is adopted but under what conditions, through which mechanisms, and with what results. By analyzing interactions between institutional settings, farmer capacities, incentive structures, and behavioral responses, the CMO perspective reveals adoption as a contingent and context-sensitive process rather than a predictable sequence. This shift in analytical lens is particularly valuable in heterogeneous environments such as small-scale agriculture, where institutional constraints, policy frameworks, and socio-psychological factors play decisive roles.

The contribution of this study lies precisely in bridging these two perspectives. On the one hand, the diffusion framework highlights patterns of spread and general drivers of adoption; on the other, the realist approach uncovers the mechanisms through which adoption succeeds or fails in specific contexts. Integrating these lenses allows us to move beyond generic accounts of "barriers" to innovation and to articulate differentiated typologies of small-scale farmers, linked to policy levers and institutional dynamics.

Figure 4 illustrates this theoretical contribution by mapping the interplay between contextual variables (e.g., access to credit, policy frameworks, extension services), mechanisms (e.g., trust, incentives, collective action), and outcomes (e.g., adoption intensity, sustainability impacts). The framework visually demonstrates how adoption is not the result of isolated decisions but of dynamic interactions between structure, agency, and environment. This synthesis represents an important step forward in conceptualizing adoption as a

multidimensional process, aligning innovation diffusion theory with realist explanations of causality.

4.4. Limitations

Despite the breadth and methodological rigor of this review, several limitations must be acknowledged. First, the geographical distribution of the studies reveals an overrepresentation of countries with consolidated research systems, such as Australia, Canada, and members of the European Union. This concentration introduces a structural bias that limits the generalizability of the findings to regions where research capacity and institutional frameworks are weaker, particularly in parts of Latin America, Africa, and Asia.

Second, although the temporal scope (2019–2024) ensured the inclusion of the most recent studies, it also excluded longer-term analyses that could provide valuable insights into historical trajectories of technology adoption. The absence of longitudinal and comparative datasets reduces the ability to capture structural dynamics and path dependencies across regions.

Third, while the realist approach (CMO) allowed us to explore context–mechanism–outcome interactions, the evidence base remains limited in addressing farmers' psychological and cognitive dimensions. Few studies incorporate behavioral economics or risk perception frameworks, which restricts the capacity to understand how attitudes, beliefs, and cognitive biases influence adoption decisions.

Fourth, the review is constrained by the heterogeneity of methodologies employed across the included studies. The diversity of designs—ranging from econometric modeling to qualitative case studies—hinders direct comparability of results, making synthesis necessarily thematic rather than statistical. These limitations directly affect the scope of the conclusions: while the review offers a robust cartography of adoption dynamics, the findings should be interpreted as context-sensitive insights rather than universally generalizable laws.

Future research should address these gaps by expanding comparative and cross-regional analyses, particularly in underrepresented areas of the Global South; integrating socio-psychological perspectives into adoption studies; and promoting longitudinal designs that capture the evolution of adoption trajectories over time. Additionally, greater methodological standardization in data collection and reporting would strengthen the comparability of evidence and support more systematic meta-analyses in the future.

5. Conclusions

By adopting a realist synthesis approach and weighing evidence epistemologically, this review has moved beyond incremental studies that merely confirm obvious truths. We have shown that while Rogers' classic Diffusion of Innovations model remains relevant, its utility is limited unless expanded to account for specific contextual factors and the inherent tensions of the Farm to Fork (F2F) Strategy [57]. Our key contribution is not only to identify barriers to adoption, but also to unpack how these barriers manifest differently across diverse typologies of small-scale farmers, and how technological adoption itself can generate paradoxes that complicate the achievement of policy objectives.

Synthesis of the Main Contribution

This systematic review has identified several limitations in the adoption of agricultural innovations, particularly within organic farming and technological systems. Although these innovations hold significant potential for improving sustainability and food security, current literature lacks comprehensive evaluations of the socio-psychological dimensions that influence farmers' decision-making processes. Chindasombatcharoen et al. (2024) emphasize that internal factors—such as trust, attitudes, and perceived effort—

can be as critical as technical or financial constraints, especially among smallholders in transitioning economies [59].

External barriers also persist. Restrepo-Campuzano et al. (2023) found that more than half of the barriers to adopting sustainable agricultural innovations are related to policy, infrastructure, and institutional support [23]. These findings highlight the need for regulatory reforms, incentive mechanisms, and farmer-centered extension systems that reduce the burden of adoption and ensure equitable access to technologies.

Moreover, although technologies such as precision agriculture and blockchain systems provide tangible improvements in efficiency and sustainability, their scalability remains uneven across regions. Weber et al. (2025) warn that without innovation-driven policies, initiatives such as the EU's Farm to Fork strategy may result in reduced productivity and greater trade dependency [61]. Similarly, Pedersen et al. (2024) call for integrated governance frameworks that align institutional, technical, and financial instruments to accelerate adoption in rural and resource-constrained settings [65]. The uncritical promotion of technology risks intensifying conflicts between environmental sustainability and short-term economic viability for certain groups of farmers—an essential finding largely underestimated in existing literature.

Future Research Agenda

This study proposes a research agenda to address the knowledge gaps identified:

- In-depth socio-psychological studies: There is a scarcity of research exploring farmers' perceptions, trust, and resistance to change, particularly in the context of F2F.
- Analysis of trade-off mechanisms: More quantitative and qualitative studies are needed to examine the precise mechanisms through which technology creates tensions among policy goals (e.g., the impact of smart irrigation systems on basin-level water availability).
- Expansion of geographic coverage: Much of the current literature focuses on developed economies. It is crucial to expand research into emerging economies exporting to the EU, to understand how F2F regulations influence technological adoption within their supply chains.

In sum, this study has demonstrated that the adoption of agricultural innovations is an inherently complex process that extends far beyond the mere availability of technology. By moving beyond Rogers' traditional Diffusion of Innovations model and adopting a Realist Review framework, we have shown that socioeconomic context and institutional barriers are decisive factors in the success or failure of adoption [57]. The findings underscore that the uncritical promotion of technologies, without considering the diverse typologies of small-scale farmers and the inherent paradoxes (such as the tension between sustainability and economic viability), can lead to unintended consequences. True agricultural transformation therefore requires a differentiated approach, where policies are designed with a deep understanding of local contexts, investing in infrastructure, training, and support systems that strengthen farmers' capacity to thrive in a changing global environment.

Limitations and Implications for Interpretation

These conclusions must be interpreted considering the study's limitations. The concentration of research in countries with consolidated infrastructures, the absence of cross-regional comparative evidence, and the underrepresentation of socio-psychological dimensions restrict the generalizability of the findings. Consequently, the insights presented here should be viewed as context-sensitive rather than universally applicable, reinforcing the need for tailored, region-specific policies and future research that broadens the empirical foundation.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su17219493/s1: Table S1: PRISMA 2020 Checklist. Reference [68] is cited in the Supplementary Materials.

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